

Mark Scheme — Practice Paper 1 (P1)

M = method mark, A = accuracy mark (depends on M), B = independent mark. FT = follow through. 教师/批改用，请勿在考试前给学生。

1 $2(x-3)^2 - 11$: B1 for $a = 2$, B1 for $b = -3$ (and $c = -11$); B1 min value -11 at $x = 3$. [3]

2 Use $\sin^2 \theta = 1 - \cos^2 \theta$: $2 - 2\cos^2 \theta + 3\cos \theta = 3$ (M1) $\Rightarrow 2\cos^2 \theta - 3\cos \theta + 1 = 0 \Rightarrow (2\cos \theta - 1)(\cos \theta - 1) = 0$ (M1). $\cos \theta = \frac{1}{2}$: $\theta = 60^\circ, 300^\circ$ (A1); $\cos \theta = 1$: $\theta = 0^\circ, 360^\circ$ (A1). Four answers: $0^\circ, 60^\circ, 300^\circ, 360^\circ$. [4]

3 Midpoint $(4, 1)$ (B1). Gradient $AB = \frac{-3-5}{7-1} = -\frac{4}{3}$, so perpendicular gradient $= \frac{3}{4}$ (M1 A1). Line: $y - 1 = \frac{3}{4}(x - 4) \Rightarrow 3x - 4y - 8 = 0$ (A1). [4]

4(a) $ar = 12$, $\frac{a}{1-r} = 64$ (B1 both). Substitute $a = 64(1-r)$: $64r(1-r) = 12 \Rightarrow 16r^2 - 16r + 3 = 0$ (M1, AG). Factorise $(4r-1)(4r-3) = 0$: $r = \frac{1}{4}$ or $r = \frac{3}{4}$ (A1). [3]

4(b) $r = \frac{3}{4}$, $a = 16$ (B1 FT). $S_5 = \frac{16(1-(\frac{3}{4})^5)}{1-\frac{3}{4}} = \frac{781}{16} = 48.8$ (3 s.f.) (M1 A1 — accept 48.8 or 48.8125). [2]

5(a) $fg(x) = 2(x^2 - 4x) + 3 = 2x^2 - 8x + 3$ (B1). $2x^2 - 8x + 3 = 13 \Rightarrow x^2 - 4x - 5 = 0 \Rightarrow (x-5)(x+1) = 0$ (M1). $x = 5$ only (reject $x = -1$ since domain of g is $x \geq 2$) (A1). [3]

5(b) $y = (x-2)^2 - 4$ (M1 complete the square or use formula), $x = 2 \pm \sqrt{y+4}$; take $+$ because $x \geq 2$: $g^{-1}(x) = 2 + \sqrt{x+4}$ (A1). Domain $x \geq -4$ (B1, = range of g). [3]

6(a) Arc $AB = 8 \times 1.2 = 9.6$ (B1). $OC = 8 \cos 1.2 = 2.899$, $AC = 8 \sin 1.2 = 7.456$ (M1). Perimeter $= 9.6 + (8 - 2.899) + 7.456 = 22.2$ cm (A1; 22.16). [3]

6(b) Sector area $= \frac{1}{2} \times 8^2 \times 1.2 = 38.4$ (B1). Triangle $OAC = \frac{1}{2} \times 2.899 \times 7.456 = 10.81$ (M1). Region $= 38.4 - 10.81 = 27.6$ cm² (A1; 27.59). [3]

7(a) $\frac{dy}{dx} = 3x^2 - 12x + 9$ (B1). $= 3(x-1)(x-3) = 0 \Rightarrow x = 1, 3$ (M1). Points $(1, 5)$ and $(3, 1)$ (A1). $\frac{d^2y}{dx^2} = 6x - 12$: at $x = 1$ negative $\Rightarrow$ maximum at $(1, 5)$; at $x = 3$ positive $\Rightarrow$ minimum at $(3, 1)$ (A1). [4]

7(b) At $x = 2$: $y = 3$, $\frac{dy}{dx} = 12 - 24 + 9 = -3$ (B1). Normal gradient $= \frac{1}{3}$ (M1). $y - 3 = \frac{1}{3}(x - 2) \Rightarrow x - 3y + 7 = 0$ (A1). [3]

8(a) $4x - x^2 = x \Rightarrow x(3-x) = 0$ (M1); $P = (3, 3)$ (A1). [2]

8(b) Area $= \int_0^3 [(4x - x^2) - x] dx = \int_0^3 (3x - x^2) dx$ (M1 subtract, M1 integrate) $= [\frac{3}{2}x^2 - \frac{1}{3}x^3]_0^3 = \frac{27}{2} - 9$ (A1) $= \frac{9}{2} = 4.5$ (A1). [4]

8(c) $V = \pi \int_0^3 (4x - x^2)^2 dx = \pi \int_0^3 (16x^2 - 8x^3 + x^4) dx$ (M1 expand & set up) $= \pi [\frac{16}{3}x^3 - 2x^4 + \frac{1}{5}x^5]_0^3$ (A1) $= \pi(144 - 162 + 48.6) = 30.6\pi$ (A1; $= \frac{153}{5}\pi$). [3]

9(a) LHS $= \frac{(1 + \sin \theta) - (1 - \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)}$ (M1) $= \frac{2 \sin \theta}{1 - \sin^2 \theta} = \frac{2 \sin \theta}{\cos^2 \theta}$ (M1, uses $\sin^2 + \cos^2 = 1$) $= \frac{2 \sin \theta}{\cos \theta} \cdot \frac{1}{\cos \theta} = \frac{2 \tan \theta}{\cos \theta}$ (A1, AG). [3]

9(b) $\frac{2 \tan \theta}{\cos \theta} = 8 \tan \theta \Rightarrow \tan \theta \left(\frac{2}{\cos \theta} - 8 \right) = 0$ (M1 — do **not** divide through by $\tan \theta$ without considering $\tan \theta = 0$). $\tan \theta = 0 \Rightarrow \theta = 180^\circ$ (A1; 0° and 360° are excluded by the strict inequality).

$\cos \theta = \frac{1}{4} \Rightarrow \theta = 75.5^\circ$ (A1), 284.5° (A1). [4]

9(c) $2x - \frac{\pi}{3}$ lies in $[-\frac{\pi}{3}, \frac{5\pi}{3}]$; $\sin = \frac{1}{2}$ at $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ (M1). $2x = \frac{\pi}{2} \Rightarrow x = \frac{\pi}{4}$ (A1); $2x = \frac{7\pi}{6} \Rightarrow x = \frac{7\pi}{12}$ (A1). [3]

10(a) $(x - 3)^2 + (y + 2)^2 = 25$: centre $(3, -2)$ (B1), radius 5 (B1). [2]

10(b) Method 1: distance from centre to line = radius: $\frac{|2(3) - (-2) + k|}{\sqrt{5}} = 5$ (M1 A1) $\Rightarrow |8 + k| = 5\sqrt{5}$ (M1) $\Rightarrow k = -8 \pm 5\sqrt{5}$ (A1). Method 2: substitute $y = 2x + k$: $5x^2 + (4k + 2)x + (k^2 + 4k - 12) = 0$ (M1 A1); discriminant = 0: $(4k + 2)^2 - 20(k^2 + 4k - 12) = 0 \Rightarrow k^2 + 16k - 61 = 0$ (M1) $\Rightarrow k = -8 \pm \sqrt{125} = -8 \pm 5\sqrt{5}$ (A1). (Decimal 3.18, -19.2 gets A0 unless exact form also given.) [4]

10(c) $49 + 1 - 42 + 4 - 12 = 0$ ✓ (B1). Gradient of radius $CP = \frac{1 - (-2)}{7 - 3} = \frac{3}{4}$ (M1); tangent gradient $-\frac{4}{3}$ (A1). $y - 1 = -\frac{4}{3}(x - 7) \Rightarrow 4x + 3y - 31 = 0$ (A1). [4]

11(a) $a + 4d = 17$ (B1); $\frac{10}{2}(2a + 9d) = 185 \Rightarrow 2a + 9d = 37$ (B1). Solve: $d = 3$, $a = 5$ (M1 A1 — award 3 marks total: B1 B1 then A1 for both values; treat solving as implied). [3]

11(b) $\frac{n}{2}(10 + 3(n - 1)) > 1000 \Rightarrow 3n^2 + 7n - 2000 > 0$ (M1 A1). Critical value $n = \frac{-7 + \sqrt{24049}}{6} = 24.7$, so least $n = 25$ (A1). Check: $S_{24} = 948$, $S_{25} = 1025$. [3]

12(a) Replace x by $x - 3$ and y by $y + 2$: $y + 2 = (x - 3)^2 - 4(x - 3) + 1$ (M1) $\Rightarrow y = x^2 - 10x + 20$ (A1). [2]

12(b) Stretch parallel to the x -axis with scale factor $\frac{1}{2}$ (B1 stretch in x , B1 factor $\frac{1}{2}$); stretch parallel to the y -axis with scale factor 3 (B1). Either order is acceptable here. [3]

Total: 75